

Milestone-C Build Restoration

Updating the maintained TestNG build path

1 Document details

Field	Recorded value
Prepared by	Darvin Anorma Mangampo
Assessment period	13–14 September 2026 (UTC+08:00)
Repository	selenium-saucedemo-remediation
Document date	14 September 2026
Technical commit	a4e5f9829ed5f6047ca5672c8bb425ccc6580be1

2 Purpose

Milestone-C started from the published Milestone-B head. I kept the existing maintainedTest source set and moved its build path to Gradle 8.14.5 under Eclipse Temurin JDK 21. The lane's own TestNG dependency changed from 7.3.0 to 7.12.0.

The work was limited to two compile-only runs of the maintained source and the Gradle results they returned. I did not compile the inherited src/test tree, open a browser or test an application function.

3 Repository origin and change boundary

The application source and legacy test framework were inherited and remained unchanged. My Milestone-C changes are limited to build.gradle and the four Gradle Wrapper files. The probe source added in Milestone-B also remained unchanged.

Boundary	Recorded position
Starting head	c5a969652ecd93546318bb95b9d389ed16a05de3
Inherited baseline	inherited-baseline-2026-09-05 at 4929e146a856d3b9ca2f04f5fda2eb014f599aa4
Protected workbook	SHA-256 9ECECF817507E4762A560DD1BFDA6C563B79648114B4121C1C54906CFDA07B9C
Permitted implementation	build.gradle gradle/wrapper/gradle-wrapper.jar gradle/wrapper/gradle-wrapper.properties gradlew gradlew.bat

4 Activity record

Activity	Work performed	Recorded result
M-C-A1	Checked the published head and workbook fingerprint. Before editing, I recorded the versions to be used and limited the work to five files.	The starting point matched Milestone-B. The version and file boundaries were documented before implementation.
M-C-A2	Prepared JDK 21 outside the repository and regenerated the wrapper. I updated only the maintained lane dependency and active repository setting.	Gradle 8.14.5 ran under JDK 21. The maintained dry graph and classpath stayed isolated.
M-C-A3	Performed two forced compile-only builds from clean output with caches disabled. Using the same source, the run with integrityProbe=fail returned one failed result. The run with integrityProbe=pass returned one passing result.	Both compiles succeeded with the same class hash. Each test run recorded one invocation.
M-C-A4	Reviewed and staged the five permitted files, then created the signed technical commit and pushed it.	The remote matched the local commit at closure. The worktree was clean.

5 Minimum build modernisation

Before extracting the Eclipse Temurin JDK 21 ZIP, I checked its SHA-256 value. I kept the JDK and Gradle user home outside the repository. I did not change the system Java setting.

The wrapper task generated the Gradle 8.14.5 launch files. I recorded the distribution checksum in `gradle-wrapper.properties` and compared the generated wrapper JAR with Gradle's published value. JCenter was removed from active dependency resolution, while Maven Central remained.

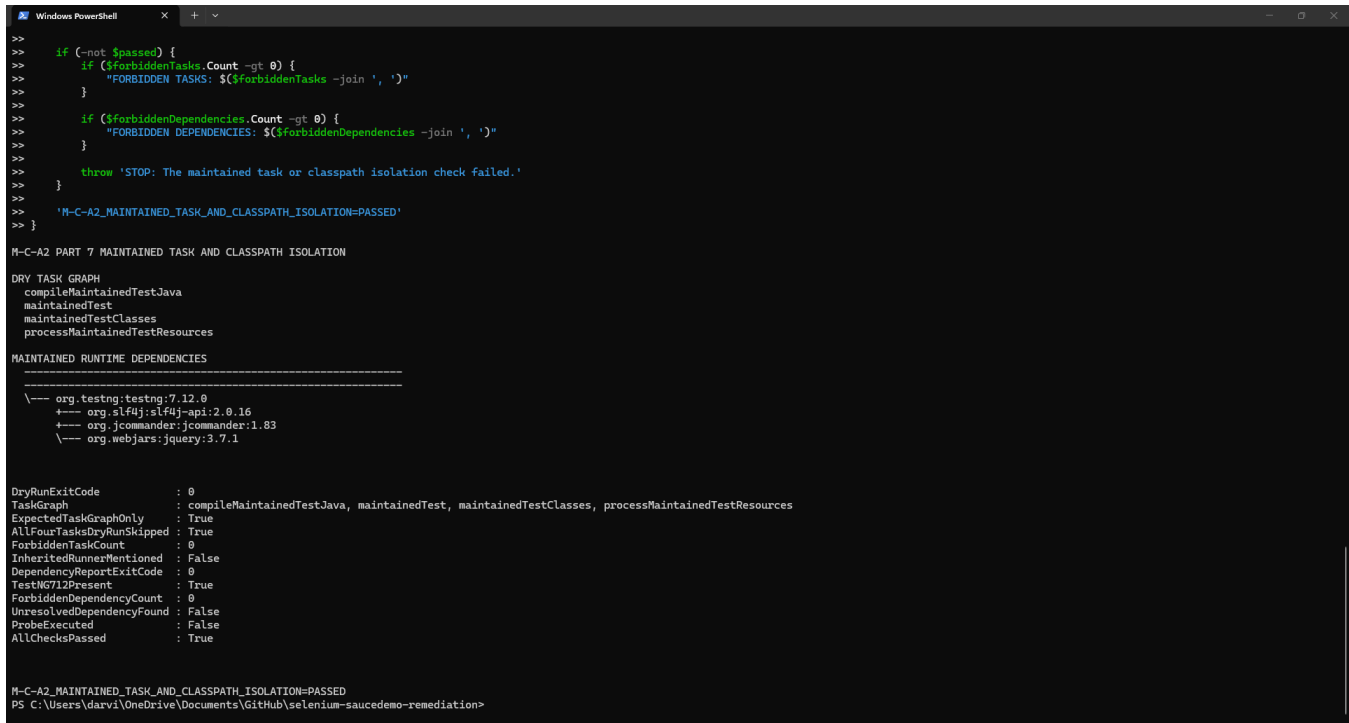
Control	Verified result
Gradle runtime	8.14.5 under Eclipse Temurin JDK 21.0.12.1+1
Maintained toolchain	Java 21 compiler and test launcher; release level 21
Gradle distribution SHA-256	6f74b601422d6d6fc4e1f9a1ab6522f642c2fdcbc15ae33ebd30ba3d7198e854
Wrapper JAR SHA-256	7d3a4ac4de1c32b59bc6a4eb8ecb8e612ccd0cf1ae1e99f66902da64df296172
Maintained TestNG	7.12.0
Dependency repository	Maven Central; JCenter removed

Allure compatibility boundary

The first Gradle 8 help run stopped while creating the inherited `allureReport` task. The error referred to `TaskReportContainer`, which the old Allure Gradle plugin expected. I removed the Allure plugin declaration and its configuration block after this failure. The inherited `allure-testng 2.13.5` dependency was left unchanged, and it did not enter the maintained runtime classpath. No other legacy integration was upgraded.

6 Maintained task and classpath isolation

The Gradle --dry-run task-graph check listed compileMaintainedTestJava, processMaintainedTestResources, maintainedTestClasses and maintainedTest without executing them. It did not include compileTestJava or an inherited test runner. The maintained runtime dependency report resolved TestNG 7.12.0 with SLF4J API 2.0.16, JCommander 1.83 and jQuery 3.7.1. Selenium, Cucumber, Allure and Rest Assured were absent from the maintained classpath.



```

>> if (-not $passed) {
>>     if ($forbiddenTasks.Count -gt 0) {
>>         "FORBIDDEN TASKS: $($forbiddenTasks -join ', ')"
>>     }
>>     if ($forbiddenDependencies.Count -gt 0) {
>>         "FORBIDDEN DEPENDENCIES: $($forbiddenDependencies -join ', ')"
>>     }
>>     throw 'STOP: The maintained task or classpath isolation check failed.'
>> }
>> 'M-C-A2_MAINTAINED_TASK_AND_CLASSPATH_ISOLATION=PASSED'
>> }

M-C-A2 PART 7 MAINTAINED TASK AND CLASSPATH ISOLATION

DRY TASK GRAPH
compileMaintainedTestJava
maintainedTest
maintainedTestClasses
processMaintainedTestResources

MAINTAINED RUNTIME DEPENDENCIES
-----
\--- org.testng:testng:7.12.0
     +--- org.slf4j:slf4j-api:2.0.16
     +--- org.jcommander:jcommander:1.83
     \--- org.webjars:jquery:3.7.1

DryRunExitCode      : 0
TaskGraph           : compileMaintainedTestJava, maintainedTest, maintainedTestClasses, processMaintainedTestResources
ExpectedTaskGraphOnly : True
AllFourTasksDryRunSkipped : True
ForbiddenTaskCount   : 0
InheritedRunnerMentioned : False
DependencyReportExitCode : 0
TestNGPresent       : True
ForbiddenDependencyCount : 0
UnresolvedDependencyFound : False
ProbeExecuted       : False
AllChecksPassed      : True

M-C-A2_MAINTAINED_TASK_AND_CLASSPATH_ISOLATION=PASSED
PS C:\Users\darvi\OneDrive\Documents\GitHub\selenium-saucedemo-remediation>

```

Figure 1. The maintained dry task graph and TestNG 7.12.0 classpath remained separate from the inherited framework.

7 Compilation results

Generated build output was cleared before the compile proof. I forced compileMaintainedTestJava to execute twice with the build cache and configuration cache disabled. Both runs used unchanged source and returned exit code 0.

Check	Run 1	Run 2	Comparison
Process result	Exit 0	Exit 0	Both runs passed
Compile task	Executed	Executed	No stale task result
Probe class SHA-256	D888EAE826670CDCE11509C9A47B91A3E8BE5B392F464BC7AB75DAABE9248790	D888EAE826670CDCE11509C9A47B91A3E8BE5B392F464BC7AB75DAABE9248790	Exact match

```

>>> 'M-C-A3_REPRODUCIBLE_MAINTAINED_COMPILATION'
>>> ..
>>>
>>> $summary = [ordered]@{
>>>     CompileRun1ExitCode = $run1.ExitCode
>>>     CompileRun1Executed = $run1.CompileExecuted
>>>     CompileRun2ExitCode = $run2.ExitCode
>>>     CompileRun2Executed = $run2.CompileExecuted
>>>     Run1ClassSHA256     = $run1.ClassSHA256
>>>     Run2ClassSHA256     = $run2.ClassSHA256
>>>     CompiledClassesMatch = $classHashesMatch
>>>     ForbiddenTaskCount  = ($run1.ForbiddenTaskCount + $run2.ForbiddenTaskCount)
>>>     SourceHashesUnchanged = $sourceUnchanged
>>>     AllChecksPassed     = $passed
>>> }
>>> [PSCustomObject]$summary | Format-List
>>>
>>> $summary |
>>>     ConvertTo-Json |
>>>     Set-Content '
>>>         -Encoding UTF8 '
>>>         -LiteralPath (Join-Path $records 'M-C-A3-Compilation-Summary.json')
>>>
>>> if (-not $passed) {
>>>     throw 'STOP: Reproducible maintained compilation was not established.'
>>> }
>>>
>>> 'M-C-A3_REPRODUCIBLE_MAINTAINED_COMPILATION=PASSED'
>>> }
M-C-A3_REPRODUCIBLE_MAINTAINED_COMPILATION

CompileRun1ExitCode : 0
CompileRun1Executed : True
CompileRun2ExitCode : 0
CompileRun2Executed : True
Run1ClassSHA256     : D888EAE826670CDCE11509C9A47B91A3E8BE5B392F464BC7AB75DAABE9248790
Run2ClassSHA256     : D888EAE826670CDCE11509C9A47B91A3E8BE5B392F464BC7AB75DAABE9248790
CompiledClassesMatch : True
ForbiddenTaskCount  : 0
SourceHashesUnchanged : True
AllChecksPassed     : True

M-C-A3_REPRODUCIBLE_MAINTAINED_COMPILATION=PASSED
PS C:\Users\darvi\OneDrive\Documents\GitHub\selenium-saucedemo-remediation>

```

Figure 2. Two forced maintained compiles executed from unchanged source and produced the same probe class hash.

8 Probe execution results

After the compile proof, I ran the existing probe once with integrityProbe=fail. I saved its console and XML result outside the repository before running integrityProbe=pass from the same source. The XML for each run contained one test case.

Check	Failure run	Passing run	Recorded meaning
Gradle exit	1	0	The process result followed the TestNG result
XML result	1 failed	1 passed	Only ExecutionIntegrityProbe ran
Invocation count	1	1	No retry was observed
Freshness	Not cached	Not cached	Both test tasks executed

```

>> ConvertTo-Json |
>> Set-Content '
>>   -Encoding UTF8 '
>>   -LiteralPath (Join-Path $records 'M-C-A3-Final-Summary.json')
>>
>> if (-not $passedAll) {
>>   throw 'STOP: M-C-A3 did not meet every exit condition.'
>> }
>>
>> 'M-C-A3_FAILURE_PASS_AND_INTEGRITY=PASSED'
>> }
warning: in the working copy of 'gradle/wrapper/gradle-wrapper.properties', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'gradlew', LF will be replaced by CRLF the next time Git touches it
M-C-A3 FAILURE, PASS AND RETAINED INTEGRITY

APPROVED WORKING CHANGES
build.gradle
gradle/wrapper/gradle-wrapper.jar
gradle/wrapper/gradle-wrapper.properties
gradlew
gradlew.bat

DependencyIsolationRetained : True
BothCompilationsPassed     : True
CompiledClassesMatch       : True
FailureExitCode             : 1
FailureCount                : 1
FailureInvocations          : 1
PassExitCode                : 0
PassCount                  : 1
PassInvocations             : 1
NoRetryObserved             : True
NoCachedTestResult          : True
InheritedOrBrowserWorkFound : False
BuildSourceUnchanged        : True
ProbeSourceUnchanged        : True
WorkbookSHA256              : 9E9ECF817597E4762A568DD1BFDA6C563879648114B4121C1C54986CFDA97B9C
WorkbookUnchanged           : True
ChangedPathCount            : 5
ApprovedScopeOnly           : True
StagedPathCount             : 0
GeneratedOutputIgnored      : True
RecordsRetained              : True
AllChecksPassed             : True

M-C-A3_FAILURE_PASS_AND_INTEGRITY=PASSED
PS C:\Users\darvi\OneDrive\Documents\GitHub\selenium-saucedemo-remediation>

```

Figure 3. The probe ran once with each outcome setting. The retained results show one invocation per run and no retry.

9 Technical change closure

The final diff contained only the five paths permitted in M-C-A1. I reviewed the text changes, verified the wrapper JAR by hash and staged those files. The signed commit was pushed to the approved origin. At the recorded closure, local HEAD and remote master pointed to the same commit, with a clean worktree and no generated or publication file in the commit.

Control	Recorded result
Technical commit	a4e5f9829ed5f6047ca5672c8bb425ccc6580be1
Parent commit	c5a969652ecd93546318bb95b9d389ed16a05de3
Commit subject	build: modernise isolated maintained TestNG lane
Committed paths	build.gradle and the four Gradle Wrapper files
Remote state at closure	Local HEAD, origin/master and remote master matched; 0 commits ahead
Build source SHA-256 (Windows working copy)	7A61DF6C59282180494408144EAFBF615A2DB3CF6D7BB9D65382A44853CB99D
Probe source SHA-256 (Windows working copy)	9BFE63F15901DF31014A939BD8538C6BDA17D1EF8E938C435CBF5633522FC6FB
Workbook SHA-256	9ECECF817507E4762A560DD1BFDA6C563B79648114B4121C1C54906CFDA07B9C

10 Limits

These findings apply only to the maintained non-browser TestNG lane. The inherited src/test tree was not compiled. No Selenium session started, and no application function ran. The failed result came from ExecutionIntegrityProbe using the fail setting, so it is not an application defect.

The inherited Allure dependency and other legacy integrations were not modernised. Removing the old Allure Gradle plugin configuration resolved the recorded root-configuration blocker. This does not show that the inherited reporting works. Milestone-C does not claim that the entire repository is restored or ready for functional regression testing.

11 Conclusion

Milestone-C restored the build path for the isolated maintainedTest lane without bringing the inherited browser framework into this work. The lane now uses Gradle 8.14.5 under JDK 21. Its own TestNG dependency is 7.12.0 and resolves from Maven Central.

Both forced compiles completed from unchanged source and produced the same class hash. When ExecutionIntegrityProbe ran with the fail setting, Gradle returned a failed process. The same probe later passed once with the pass setting. Each retained XML result contained one invocation, and no retry was observed.

The final commit stayed within the five approved build and wrapper paths. The workbook and probe source kept their recorded hashes. This closes build restoration for the maintained non-browser lane. Selenium execution and application testing remain outside this result and belong to later approved milestones.

12 Repository references

- [Milestone-C signed technical commit](#)
- [build.gradle at the Milestone-C commit](#)
- [Gradle Wrapper properties at the Milestone-C commit](#)
- [ExecutionIntegrityProbe source retained at the Milestone-C commit](#)
- [Published Milestone-B starting commit](#)
- [Inherited baseline tag](#)
- [Gradle Java compatibility](#)
- [Gradle Wrapper and checksum guidance](#)
- [TestNG project requirements](#)